



The business problem comes first

How can the store reduce peak-hour queue time while protecting service, cost, and risk?

THE PROBLEM

The queue is too long.

- Average peak-hour queue: 7.5 minutes.
- The store handles 84 transactions per hour.
- Customers still need accessible service.

DO NOT JUMP TO ONE ANSWER

Several methods may help.

- Keep a staffed lane open.
- Add clearer kiosk instructions.
- Place an employee near the kiosks.
- Test two kiosks in a small pilot.

HOW COT HELPS

Compare methods before choosing one.

- Separate facts from assumptions.
- Use the same criteria for each method.
- Test the most promising option before scaling.

What CoT does in this class

CoT is a structured decision rationale. It helps students inspect the basis for a recommendation.

START WITH A QUESTION

State the business problem.

- What must improve?
- What constraints matter?
- What counts as success?

ASK FOR A STRUCTURE

Make the decision logic visible.

- Facts provided.
- Assumptions to test.
- Options and trade-offs.

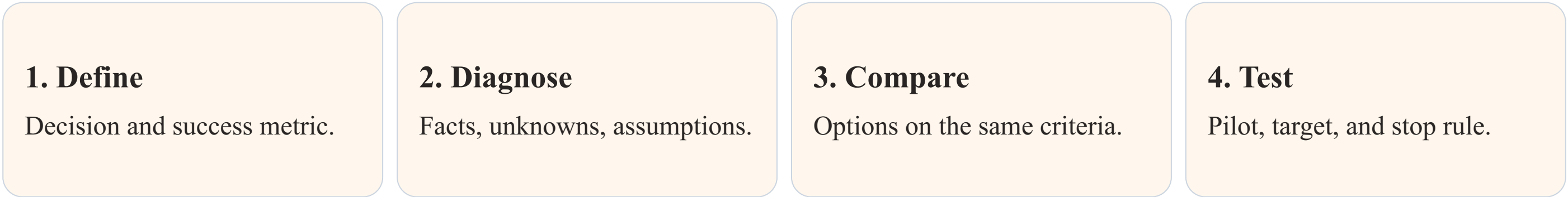
USE THE RESULT CAREFULLY

CoT is not proof.

- Check facts against the dataset.
- Ask for missing information.
- Test before scaling.

Use a four-step decision frame

The model should organize a decision. It should not silently invent the brief.



Bad prompt: no decision frame

The question sounds clear, but it does not define the business problem.

PROMPT TO AVOID

Should our grocery store
replace cashiers with
self-checkout kiosks?

WHAT IS MISSING?

- Store facts and budget.
- A success metric.
- A limit on assumptions.

Bad prompt: false certainty

Do not ask the model to guarantee a forecast that your data cannot support.

PROMPT TO AVOID

We have 12 employees.
Tell us exactly how much
profit two kiosks will add.
Guarantee customer acceptance.

WHY IT FAILS

- No cost or traffic data.
- Guarantees are impossible.
- It rewards invented numbers.

Good prompt: diagnose before deciding

Copy this prompt when you need to turn an unclear question into a research plan.

COPY AND ADAPT

Supermarket case

Known: 12 employees

Goal: shorter peak queues

Risks: theft, downtime, access

Make three lists:

1. Facts
2. Assumptions to test
3. Missing data

Do not invent cost estimates.

How to audit the response

Before accepting the output, ask whether every claim has a place in your evidence.

CHECK 1

Facts

- Can you point to a source for each fact?

CHECK 2

Assumptions

- Can each assumption be tested before rollout?

CHECK 3

Unknowns

- Did the answer clearly request missing data?

Session 1.4 self-checkout pilot data

Use this fictional instructional dataset in the next prompt and exercises. The CSV file contains the same values.

DEMAND

Peak period

- 84 transactions per hour.
- 7.5-minute average queue.
- 20% of shoppers aged 65 or older.

COST

Proposed pilot

- Two self-checkout kiosks.
- \$1,150 monthly lease per kiosk.
- \$250 monthly support cost.

RISK AND TARGET

90-day guardrails

- 1.2% baseline shrinkage.
- Target: 5-minute queue and 98% uptime.
- Stop at shrinkage above 1.4%.

Good prompt: compare options

The data are already provided. Do not invent values that are absent from the case.

COPY AND ADAPT: PART 1

Use only the Session 1.4
self-checkout pilot data.

Compare three options:

- current checkout
- pilot two kiosks for 90 days
- expand all lanes

Do not invent missing values.

Good prompt: make the pilot decision

This second section turns the comparison into an action that can be stopped or scaled.

COPY AND ADAPT: PART 2

Use these criteria:

- customer wait time
- total operating cost
- shrinkage risk
- staff workload
- service access

Recommend one option.

State one pilot target
and one stop rule.

Say "Information needed"
when evidence is missing.

Exercise 1: Assumption audit

Use the data on Slide 7 and the CSV file. Submit one audit table, not a generic recommendation.

STEP 1

Write the brief

- Decision owner.
- Deadline.
- Success metric.

STEP 2

Run Good Example 1

- List facts.
- Mark assumptions.
- List unknowns.

DELIVERABLE

One-page audit

- Five questions for the owner.
- One source for each fact.

Exercise 2: Simple pilot review

After 90 days, the average peak-hour queue is 6.5 minutes. The target was five minutes or less.

STEP 1

Name the problem

- Queue time is above target.
- Customers need help at busy times.

STEP 2

Choose one action

- Place one employee near kiosks.
- Add a short instruction sign.
- Keep one staffed lane open.

STEP 3

Measure again

- Track queue time for two weeks.
- Target: five minutes or less.
- State whether the action worked.

Before you accept an AI recommendation

A recommendation is useful only when another person can check its evidence and conditions.

FRAME

Define success

- Owner, deadline, and metric.

PROTECT

Label uncertainty

- Facts, assumptions, and unknowns.

TEST

Pilot before scale

- Target, stop rule, and review owner.